DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY

CHE 331 QUIZ #3

This is a 30 minute closed-book quiz. Write your name and answer the quiz problems related to your experiment. Turn in this question sheet with your answer.

Name: ___________________________

Membrane Air Separation

Problem 1 (40%) Sketch a clearly-labeled schematic layout of two membrane separation modules connected in series, both with bore feeds. Be sure to indicate all control devices and instrumentation.

Problem 2 (40%) (a, 10%) Define "stage cut." **(b, 15%)** Explain in your own words why a high stage cut creates a large flowrate of a low-purity permeate stream. **(c, 15%)** Explain in your own words why a low stage cut creates a small flowrate of a higher-purity permeate stream.

Problem 3 (20%) Figure 1 shows the relative permeabilities of common gases and vapors in a silicon membrane. With this information in mind, explain whether a silicon membrane would be effective for the separation of the following mixed binary gas streams into their individual components: **(a, 10%)** CO/He **(b, 10%)** H_2S/CO_2.

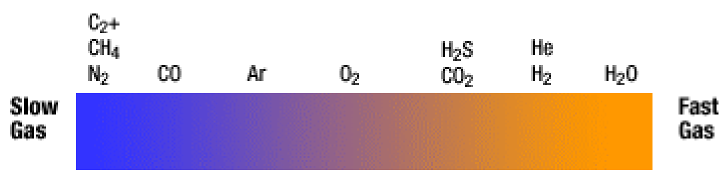

Figure 1. Relative permeabilities of some common gases and vapor in a silicon membrane [from W. L. Robb, "Thin silicone membranes--their permeation properties and some applications," *Ann. N. Y. Acad. Sci.,* 146, 119–137 (1968).]